

The Most Complex Formulas in Mathematics

A survey of advanced identities across analysis, algebra, number theory, and physics

I Analysis & Special Functions

- 1. General Leibniz / Faà di Bruno's Formula** (chain rule for higher derivatives):

$$\frac{d^n}{dx^n} f(g(x)) = \sum_{k=1}^n f^{(k)}(g(x)) \cdot B_{n,k}(g'(x), g''(x), \dots, g^{(n-k+1)}(x))$$

where $B_{n,k}$ are the (exponential) Bell polynomials.

- 2. Riemann's Explicit Formula** for the prime-counting function, relating primes to zeta zeros ρ :

$$\psi_0(x) = x - \sum_{\rho} \frac{x^{\rho}}{\rho} - \log(2\pi) - \frac{1}{2} \log(1 - x^{-2})$$

- 3. Selberg–Ramanujan-type Asymptotic Expansion** of the partition function $p(n)$ (Hardy–Ramanujan–Rademacher):

$$p(n) = \frac{1}{\pi\sqrt{2}} \sum_{k=1}^{\infty} A_k(n) \sqrt{k} \frac{d}{dn} \left(\frac{\sinh\left(\frac{\pi}{k} \sqrt{\frac{2}{3} \left(n - \frac{1}{24}\right)}\right)}{\sqrt{n - \frac{1}{24}}} \right)$$

with $A_k(n) = \sum_{0 \leq m < k, \gcd(m,k)=1} e^{\pi i s(m,k) - 2\pi i n m/k}$, and $s(m,k)$ the Dedekind sum.

- 4. Ramanujan's Master Theorem:**

$$\int_0^{\infty} x^{s-1} \left(\sum_{k=0}^{\infty} \frac{\varphi(k)}{k!} (-x)^k \right) dx = \Gamma(s) \varphi(-s)$$

- 5. Functional Equation of the Riemann Zeta Function:**

$$\zeta(s) = 2^s \pi^{s-1} \sin\left(\frac{\pi s}{2}\right) \Gamma(1-s) \zeta(1-s)$$

II Algebra, Number Theory & Modular Forms

6. Modular Discriminant / Ramanujan Tau via the Dedekind Eta Function:

$$\Delta(\tau) = (2\pi)^{12} \eta(\tau)^{24} = (2\pi)^{12} q \prod_{n=1}^{\infty} (1 - q^n)^{24}, \quad q = e^{2\pi i \tau}$$

$$\Delta(\tau) = (2\pi)^{12} \sum_{n=1}^{\infty} \tau(n) q^n$$

7. Weil Conjectures — Zeta Function of a Variety over a Finite Field:

$$Z(X, t) = \exp \left(\sum_{r=1}^{\infty} \frac{\#X(\mathbb{F}_{q^r})}{r} t^r \right) = \frac{\prod_{i=1}^{2n} P_i(t)^{(-1)^{i+1}}}{1}, \quad P_i(t) = \prod_j (1 - \alpha_{ij} t)$$

8. Birch and Swinnerton-Dyer Conjecture (analytic rank formula, conjectural):

$$L(E, s) \sim c(s-1)^r \text{ as } s \rightarrow 1, \quad r = \text{rank}(E(\mathbb{Q}))$$

$$\lim_{s \rightarrow 1} \frac{L(E, s)}{(s-1)^r} = \frac{\Omega_E \cdot \text{Reg}(E) \cdot \prod_p c_p \cdot \#\text{Sha}(E)}{\#E(\mathbb{Q})_{\text{tors}}^2}$$

9. Class Number Formula for a number field K :

$$\lim_{s \rightarrow 1} (s-1) \zeta_K(s) = \frac{2^{r_1} (2\pi)^{r_2} h_K \text{Reg}_K}{w_K \sqrt{|d_K|}}$$

10. Langlands–Kottwitz / Trace Formula (schematic form of the Arthur–Selberg trace formula):

$$\sum_{\gamma \in \Gamma} \int_{G_{\gamma} \backslash G} f(x^{-1} \gamma x) dx = \sum_{\pi} m(\pi) \text{tr } \pi(f)$$

III Geometry, Topology & Index Theory

- 11. Atiyah–Singer Index Theorem** (for an elliptic operator D on a compact manifold M):

$$\text{ind}(D) = (-1)^n \int_M \frac{\text{ch}(\sigma(D)) \text{Td}(TM \otimes \mathbb{C})}{e(TM)}$$

Special case, Hirzebruch signature theorem:

$$\text{sign}(M) = \int_M L(p_1, \dots, p_n)$$

- 12. Chern–Gauss–Bonnet Theorem** (generalized to $2n$ -dimensional manifolds):

$$\chi(M) = \frac{1}{(2\pi)^n} \int_M \text{Pf}(\Omega)$$

where Ω is the curvature 2-form and Pf the Pfaffian.

- 13. Ricci Flow Equation** (Perelman’s entropy functional, used in the Poincaré conjecture proof):

$$\begin{aligned} \frac{\partial g_{ij}}{\partial t} &= -2R_{ij}, & \mathcal{F}(g, f) &= \int_M (R + |\nabla f|^2) e^{-f} dV \\ \mathcal{W}(g, f, \tau) &= \int_M [\tau (R + |\nabla f|^2) + f - n] (4\pi\tau)^{-n/2} e^{-f} dV \end{aligned}$$

- 14. Seiberg–Witten Equations** on a 4-manifold:

$$F_A^+ = \sigma(\Phi), \quad D_A \Phi = 0$$

- 15. Yang–Mills Functional and Equations:**

$$\mathcal{S}(A) = \frac{1}{4} \int_M \text{tr}(F_A \wedge \star F_A), \quad d_A \star F_A = 0$$

IV Mathematical Physics & PDEs

16. Einstein Field Equations (general relativity):

$$R_{\mu\nu} - \frac{1}{2}R g_{\mu\nu} + \Lambda g_{\mu\nu} = \frac{8\pi G}{c^4}T_{\mu\nu}$$

17. Path Integral Formulation of Quantum Field Theory (generating functional):

$$Z[J] = \int \mathcal{D}\phi \exp\left(\frac{i}{\hbar} \int d^4x [\mathcal{L}(\phi, \partial\phi) + J(x)\phi(x)]\right)$$

18. Navier–Stokes Equations (incompressible, one of the Millennium Prize Problems):

$$\rho \left(\frac{\partial \mathbf{u}}{\partial t} + (\mathbf{u} \cdot \nabla) \mathbf{u} \right) = -\nabla p + \mu \nabla^2 \mathbf{u} + \mathbf{f}, \quad \nabla \cdot \mathbf{u} = 0$$

19. Yang–Mills Mass Gap / Wilson Loop Expectation (Millennium Prize formulation):

$$\langle W(C) \rangle = \left\langle \text{tr } \mathcal{P} \exp \left(i \oint_C A_\mu dx^\mu \right) \right\rangle \sim e^{-\sigma \text{Area}(C)}$$

20. Kardar–Parisi–Zhang (KPZ) Stochastic PDE:

$$\frac{\partial h}{\partial t} = \nu \nabla^2 h + \frac{\lambda}{2} (\nabla h)^2 + \eta(x, t)$$

V Combinatorics, Probability & Miscellany

21. Hardy–Littlewood Circle Method core identity (Vinogradov’s estimate for Goldbach-type problems):

$$r(N) = \int_0^1 S(\alpha)^3 e(-N\alpha) d\alpha, \quad S(\alpha) = \sum_{p \leq N} e(p\alpha)$$

22. Feynman–Kac Formula (connecting stochastic processes and PDEs):

$$u(x, t) = \mathbb{E} \left[e^{-\int_0^t V(X_s) ds} \varphi(X_t) \mid X_0 = x \right], \quad \frac{\partial u}{\partial t} = \frac{1}{2} \sigma^2 \frac{\partial^2 u}{\partial x^2} - V(x)u$$

23. Kolmogorov–Arnold–Moser (KAM) Frequency Condition (small-divisor problem in dynamical systems):

$$|\langle k, \omega \rangle| \geq \frac{\gamma}{|k|^\tau} \quad \text{for all } k \in \mathbb{Z}^n \setminus \{0\}$$

24. Onsager’s Exact Solution of the 2D Ising Model (free energy per site):

$$-\beta f = \log 2 + \frac{1}{8\pi^2} \int_0^{2\pi} \int_0^{2\pi} \log \left[\cosh(2\beta J)^2 - \sinh(2\beta J)(\cos \theta_1 + \cos \theta_2) \right] d\theta_1 d\theta_2$$

25. Monstrous Moonshine (McKay–Thompson series linking the Monster group to the j -invariant):

$$j(\tau) - 744 = \sum_{g \in \mathbb{M}} \frac{1}{|C(g)|} \sum_{h \in C(g)} T_g(h; \tau), \quad j(\tau) = q^{-1} + 744 + 196884q + 21493760q^2 + \dots$$

Compiled as a survey of advanced mathematical identities. Each formula spans decades of research and, in several cases (BSD, Yang–Mills, Navier–Stokes), remains the subject of open Millennium Prize problems.